Held-out DCI-I across all datasets

Gemma-2-2B

Dataset	SSAE L1	SSAE Step-L0	SSAE sigmoid-L0	SSAE ReLU	SSAE JumpReLU	SSAE TopK	SAE ReLU	SAE JumpReLU	SAE TopK
ENG→FR	0.917 ± 0.012	0.975 ± 0.000	0.975 ± 0.000	0.500 ± 0.054	0.975 ± 0.000	0.925 ± 0.071	0.650 ± 0.054	0.917 ± 0.051	0.858 ± 0.085
ENG→DE	0.754 ± 0.087	0.974 ± 0.000	0.974 ± 0.000	0.482 ± 0.097	0.956 ± 0.025	0.965 ± 0.012	0.623 ± 0.087	0.947 ± 0.021	0.956 ± 0.025
GENDER	0.958 ± 0.059	1.000 ± 0.000	1.000 ± 0.000	0.550 ± 0.035	1.000 ± 0.000	0.992 ± 0.012	0.483 ± 0.131	1.000 ± 0.000	0.992 ± 0.012
BINARY	0.680 ± 0.020	0.852 ± 0.009	0.904 ± 0.016	0.437 ± 0.026	0.840 ± 0.018	0.869 ± 0.007	0.343 ± 0.023	0.845 ± 0.008	0.843 ± 0.015
CORR	0.829 ± 0.039	0.620 ± 0.061	0.938 ± 0.022	0.472 ± 0.012	0.880 ± 0.016	0.865 ± 0.055	0.457 ± 0.016	0.882 ± 0.029	0.889 ± 0.015
TruthfulQA	0.738 ± 0.070	0.787 ± 0.028	0.894 ± 0.010	0.514 ± 0.022	0.870 ± 0.033	0.801 ± 0.029	0.807 ± 0.029	0.882 ± 0.017	0.890 ± 0.010
Sycophancy	0.141 ± 0.056	0.136 ± 0.019	0.108 ± 0.009	0.288 ± 0.081	0.166 ± 0.049	0.200 ± 0.028	0.110 ± 0.021	0.169 ± 0.065	0.134 ± 0.062
Refusal	0.620 ± 0.145	0.533 ± 0.041	0.989 ± 0.009	0.442 ± 0.067	0.971 ± 0.010	0.913 ± 0.047	0.960 ± 0.005	0.971 ± 0.005	0.993 ± 0.005
Bias-in-Bios	0.950 ± 0.000	0.949 ± 0.000	0.948 ± 0.000	0.944 ± 0.000	0.944 ± 0.001	0.957 ± 0.001	0.954 ± 0.000	0.952 ± 0.000	0.963 ± 0.000

Pythia-70M

Dataset	SSAE L1	SSAE Step-L0	SSAE sigmoid-L0	SSAE ReLU	SSAE JumpReLU	SSAE TopK	SAE ReLU	SAE JumpReLU	SAE TopK
ENG→FR	0.625 ± 0.082	0.867 ± 0.085	0.908 ± 0.024	0.500 ± 0.000	0.900 ± 0.035	0.800 ± 0.061	0.492 ± 0.012	0.825 ± 0.054	0.867 ± 0.042
ENG→DE	0.491 ± 0.054	0.807 ± 0.130	0.886 ± 0.025	0.518 ± 0.033	0.904 ± 0.025	0.798 ± 0.069	0.535 ± 0.033	0.877 ± 0.045	0.860 ± 0.087
GENDER	0.525 ± 0.035	0.817 ± 0.112	0.925 ± 0.020	0.642 ± 0.012	0.883 ± 0.047	0.842 ± 0.031	0.517 ± 0.024	0.875 ± 0.094	0.858 ± 0.082
BINARY	0.405 ± 0.010	0.485 ± 0.019	0.829 ± 0.020	0.468 ± 0.030	0.772 ± 0.008	0.552 ± 0.031	0.343 ± 0.059	0.725 ± 0.024	0.630 ± 0.036
CORR	0.524 ± 0.020	0.513 ± 0.045	0.778 ± 0.018	0.500 ± 0.000	0.656 ± 0.016	0.502 ± 0.035	0.528 ± 0.008	0.703 ± 0.041	0.677 ± 0.025
TruthfulQA	0.709 ± 0.013	0.673 ± 0.059	0.772 ± 0.010	0.654 ± 0.024	0.785 ± 0.019	0.677 ± 0.018	0.720 ± 0.026	0.770 ± 0.020	0.724 ± 0.021
Sycophancy	0.085 ± 0.007	0.057 ± 0.007	0.099 ± 0.005	0.071 ± 0.014	0.106 ± 0.016	0.146 ± 0.044	0.117 ± 0.047	0.091 ± 0.002	0.108 ± 0.004
Refusal	0.764 ± 0.074	0.504 ± 0.051	0.768 ± 0.049	0.486 ± 0.072	0.772 ± 0.009	0.685 ± 0.023	0.717 ± 0.023	0.692 ± 0.031	0.757 ± 0.085
Bias-in-Bios	0.936 ± 0.000	0.934 ± 0.000	0.939 ± 0.000	0.940 ± 0.000	0.938 ± 0.000	0.940 ± 0.000	0.945 ± 0.000	0.938 ± 0.000	0.945 ± 0.000